

To the Editorial Office
Algorithms (MDPI)

Manuscript: *DT-GSK: Dimension-Tiered Adaptive Configuration Selection and Deterministic Refinement for Gaining-Sharing Knowledge Optimization* — algorithms-4507562, revision 1

Authors: Mostafa Elsayed Masoud, Heba Sayed Mohamed Roshdy, Ali Wagdy Mohamed

Corresponding author: Mostafa Elsayed Masoud (moustafa.masoud@gmail.com)

Dear Editors,

We are pleased to submit the revised version of our manuscript, *DT-GSK: Dimension-Tiered Adaptive Configuration Selection and Deterministic Refinement for Gaining-Sharing Knowledge Optimization* (algorithms-4507562), for consideration in *Algorithms*. The point-by-point response to both reviewers accompanies this letter.

To our knowledge, DT-GSK attains the best overall CEC2017 Friedman mean rank on the seven-algorithm GSK-family panel (2.48, the unweighted mean of the four per-dimension ranks — a descriptive aggregate; eGSK is second at 2.96; the two are never Nemenyi-separable at any CEC2017 dimension; DT-GSK is second behind eGSK at $D = 30$ and on CEC2011; the CEC2011 loss is Holm-significant, and CEC2017 was configuration-selection exposed), evaluated under a release-locked protocol, with byte-stable determinism for DT-GSK in the declared supported environment. The evaluation spans five suites: alongside CEC2017, CEC2011 and the CEC2013 second comparison suite, the study reports CEC2020 under a statistical analysis plan committed to the repository before any CEC2020 result existed, and a family-internal CEC2013LSGO panel at $D = 1000$. We report the unfavorable outcomes with the favorable ones: on CEC2020 — the competition in which our comparator AGSK was the runner-up — DT-GSK places fourth of seven, consistent with the boundary condition the registered directional expectation predicted for a method whose dimension-gated subsystems are inactive at $D \leq 20$; on CEC2013LSGO, DT-GSK and AGSK share the best descriptive rank and paired tests do not separate them. The paper makes three contributions. First, a **deterministic, RNG-free final polish** executed once in the final budget slice; no convergence guarantee is claimed. The round-one revision isolated the basis this stage searches along, at the reviewers' request: the polish outperforms no refinement at both dimensions where it is active, while the learned eigenbasis it consumes is outperformed by the plain coordinate axes at $D = 50$ and not separated from them at $D = 100$, so this contribution is now claimed basis-neutrally and the eigenbasis is retained as a specified negative result. Second, a **dimension-tiered adaptive scaffold** — a bandit-style operator-configuration selector with acceptance-gated arm pruning, a tier-floored population-size reduction schedule, a hard-capped stagnation escape, and a deep-stall restart that preserves the global best — presented as an honestly labeled modified/original composite over published GSK-family mechanisms, never as a new base operator. Third, the **evaluation-integrity infrastructure** that makes the study checkable: a 13-substream, prefix-locked random-number layer, a paired optimizer-independent seed schedule, a hash-frozen configuration, and promoted, read-only evidence releases to which every reported number is bound. A supporting mechanism — the **interaction-structure memory**, a decaying, confidence- and evidence-gated coordinate-pair graph accumulated solely from strictly improving accepted moves at no extra objective

evaluations — supplies the polish eigenbasis and the linkage blocks of the block crossover at dimensions of 50 and above; its direct isolation is reported transparently as a controlled negative result — no significant standalone benefit at its active tiers, and, in the revision’s basis experiment, active harm in its only fitness-affecting channel — and we do not claim it recovers the objective’s separability structure. All comparative claims in the manuscript are scoped to this GSK-family panel; we make no field-wide performance claims.

We believe this fits *Algorithms* well: the manuscript is an algorithmic contribution with a fully specified mechanism, a reproducible artifact chain, and panel-scoped empirical evidence. In the same spirit, the paper includes a direct component isolation of the interaction-structure memory and reports its outcome transparently: the isolation finds no detectable standalone benefit from the memory at its active tiers, and we are careful not to over-attribute performance to any mechanism the isolation does not separate — at submission the deterministic compass endgame was evaluated only as a whole, and the added value of its learned basis remained unresolved. The round-one revision resolved it, and against us: on a direct three-arm contrast the coordinate axes outperform the learned eigenframe at $D = 50$. That result is now in the paper.

We confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have read and approved the submitted version. For transparency, we note that A.W.M. is the originator of the baseline GSK algorithm and a co-author of several of the family variants used as comparators, one of which (eGSK) is also co-authored by H.S.M.R.; these relationships are declared in the manuscript’s conflicts-of-interest statement. During the preparation and revision of this manuscript, the authors used two generative-AI assistants, Claude (Opus 4.6, 4.8 and 5.0; Anthropic) and ChatGPT (OpenAI), for language editing and rephrasing, for the drafting of expository prose restating findings the authors had already established from the frozen evidence, for structural review, consistency checking and review of the statistical and methodological descriptions, and for software-engineering support during implementation and tooling work, in accordance with the MDPI policy on the use of generative artificial intelligence. The algorithm design and the experimental protocol are the authors’ own, and every reported number was produced by the authors’ deterministic analysis pipeline from a version-locked evidence archive: no AI system designed an experiment, produced data, computed a statistic, or generated a scientific claim, result, or conclusion, and no AI system is an author of this work. The authors have reviewed, verified, and edited all AI-assisted text and take full responsibility for the content of this publication.

Both reviewers asked for the same thing in different words: isolate the contribution the title is named after. We ran four pre-registered experiments to do so, and two of them returned results against our submitted claims. Those results are in the revised paper, stated in the reviewers’ own terms. Thank you for your consideration.

Sincerely,

Mostafa Elsayed Masoud (corresponding author)
Heba Sayed Mohamed Roshdy
Ali Wagdy Mohamed